

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: BTECH

Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester : VII

Subject Code & Name: BTECPC703 DevOps (Development & Operations)

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. Objective type questions. (Compulsory Question)			12
1	Which of the following is NOT a core principle of DevOps? a. Collaboration b. Automation c. Isolation d. Continuous improvement	L2	1
2	What is the main goal of the DevOps delivery pipeline? a. Code obfuscation b. Continuous integration and delivery c. Manual testing d. Centralized deployment	L1	1
3	Which of these tools is commonly used in the DevOps lifecycle? - a. Docker b. Jenkins c. Kubernetes d. All of the above	L1	1
4	What does CI stand for in DevOps? a. Continuous Integration b. Cloud Infrastructure c. Collaborative Innovation d. Code Injection	L2	1
5	A DevOps engineer is expected to possess which of the following skills?	L3	1

	a. Network configuration	b. Software development	c. Infrastructure automation	d. All of the above		
6	In GIT, which command is used to view the commit history?				L2	1
	a. git commit	b. git log	c. git status	d. git push		
7	Why is DevOps needed on the cloud?				L1	1
	a. To ensure scalability of applications	b. To simplify collaboration across teams	c. To enhance automation and monitoring	d. All of the above		
8	Which of the following is a popular cloud service provider?				L1	1
	a. Amazon Web Services (AWS)	b. Azure	c. Google Cloud Platform (GCP)	d. All of the above		
9	In the context of CAMS, "blamelessness" refers to:				L2	1
	a. Avoiding accountability	b. Encouraging organizational learning by avoiding finger-pointing	c. Reducing automation failures	d. Ignoring root cause analysis		
10	What does "CAMS" stand for in DevOps?				L1	1
	a. Culture, Automation, Measurement, Sharing	b. Configuration, Automation, Metrics, Scaling	c. Collaboration, Agile, Metrics, Security	d. Code, Analysis, Monitoring, Scaling		
11	What is Docker primarily used for?				L2	1
	a. Operating system installation	b. Containerization of applications	c. Virtual machine management	d. Network configuration		
12	Which of the following is a key component of Kubernetes?				L2	1
	a. Pods	b. Containers	c. Repositories	d. Virtual machines		

Q. 2	Solve the following.		12
A)	Explain the core principles of DevOps in detail. How do these principles contribute to improving software delivery processes?	L2	6
B)	What are Continuous Integration (CI) and Continuous Delivery (CD)? Highlight their significance in the DevOps lifecycle.	L1	6
Q.3	Solve the following.		12
A)	What is cloud computing? Discuss its role in modern IT infrastructure and its relevance to DevOps practices.	L2	6
B)	How do cloud service providers like AWS, Azure, and Google Cloud enable and enhance DevOps practices?	L3	6
Q.4	Solve Any Two of the following.		12
A)	What is version control, and why is it essential in modern software development?	L2	6
B)	Explain the importance of Git in the industry. Discuss its benefits and how it supports collaborative software development.	L2	6
C)	Discuss commonly used Git commands for managing repositories, such as clone, add, commit, and push.	L1	6
Q.5	Solve Any Two of the following.		12
A)	Discuss the role of automation in DevOps. How does continuous delivery automation help deliver high value?	L2	6
B)	What is the significance of sharing in the CAMS framework? Discuss how sharing knowledge and practices improves team collaboration and learning.	L1	6
C)	What is configuration management in DevOps? Discuss its role in ensuring consistency and efficiency in software environments.	L2	6
Q.6	Solve Any Two of the following.		12

A)	Explain the history of software engineering. How has the field evolved from early programming to modern software development practices?	L1	6
B)	Describe the role of Jenkins in Continuous Integration (CI). How does Jenkins automate the software build and deployment process?	L2	6
C)	Explain how Docker, Kubernetes, Jenkins, and Ansible work together in a DevOps pipeline. Discuss their interactions and how they contribute to a streamlined deployment process.	L2	6
*** End ***			